

Quantum Mechanics

Lecture 2: Vector Spaces, Superposition, and the First Quantum Postulates

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Chapter 1

Vector Spaces, Superposition, and the First Quantum Postulates

We ended the previous lecture with a question about the uncertainty principle. Before building any new formalism, let us settle it. The apparent loophole is instructive because it distinguishes the momentum a particle had before a measurement from the momentum it has after the measurement.

1.1 Which Velocity Have We Measured?

Suppose a first position measurement locates a particle only within a region of width δ . Wait a time T , make a second measurement with the same resolution, and let the centers of the two regions be separated by L . The inferred average velocity is

$$v_{\text{avg}} \simeq \frac{L}{T}, \quad \Delta v_{\text{avg}} \sim \frac{\delta}{T}. \quad (1.1)$$

Numerical factors in the second estimate are unimportant. By making T large, we can make δ/T as small as we please. Have we measured position to accuracy δ and velocity almost exactly?

Question from the audience. Can two imprecise position measurements, separated by a very long time, evade the uncertainty principle? ^a

Response. No. Equation (1.1) gives the velocity after the first position measurement. To localize the particle on the scale δ , the probe must contain wavelengths no larger than that scale. A photon capable of doing so carries momentum of order $\hbar/\delta$, and its scattering gives the particle a kick of the same order. The later trajectory therefore reveals the momentum of the disturbed state, not the momentum immediately before localization.

^aLecture timestamp: 00:00:17–00:04:17.

Trying to verify the result does not restore the missing information. Repeating the procedure changes the momentum again by an amount of order $\hbar/\delta$. Classical physics permits an ideal limit in which position is observed ever more gently while the velocity is left alone. Quantum mechanics does not. This is the point of the uncertainty principle, and with it settled we can begin the mathematics needed for states.

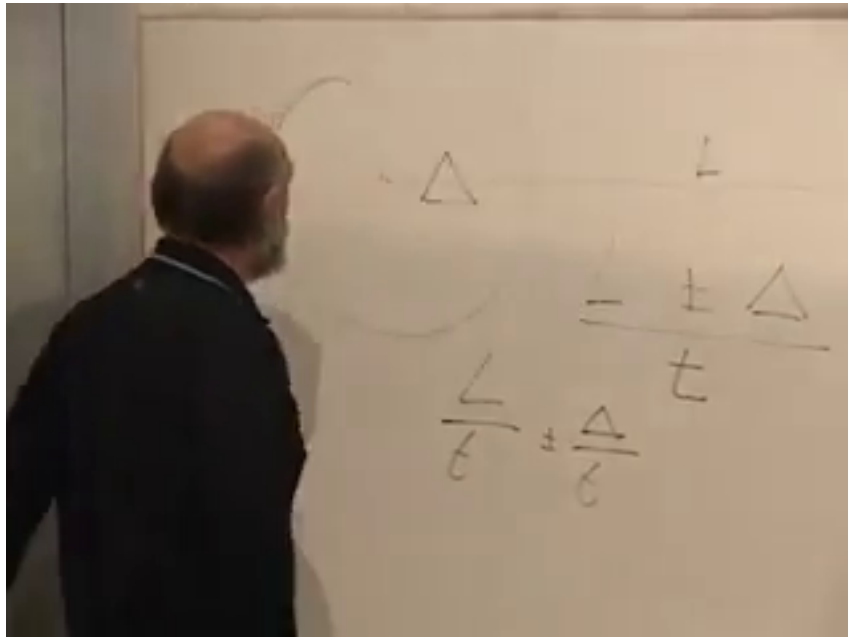


Figure 1.1: Two coarse position readings separated by a long time. The board displays the displacement L , the resolution scale δ , and the velocity uncertainty of order δ/T .

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1.2 The One-Dimensional Model: Complex Numbers

The next subject is dry, but it is not difficult. We need a quick review of complex numbers only to establish notation. Write

$$z = x + iy, \quad z^* = x - iy, \quad i^2 = -1. \quad (1.2)$$

In the complex plane, conjugation reflects a point across the real axis. Complex numbers can be added and multiplied by arbitrary complex scalars, so $\mathbb{C}$ is itself a one-dimensional complex vector space. It is the simplest model of the structures we shall use.

In this one-dimensional model, regard z as a ket and z^* as its corresponding bra. Complex conjugation exchanges the two. If

$$z_1 = x_1 + iy_1, \quad z_2 = x_2 + iy_2,$$

then

$$z_1 z_2 = (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + y_1 x_2), \quad (1.3)$$

$$(z_1 z_2)^* = z_1^* z_2^*. \quad (1.4)$$

For scalar numbers the factors commute. It is nevertheless useful to write the related identity

$$(z_1^* z_2)^* = z_2^* z_1, \quad (1.5)$$

because the reversal of bra and ket will survive when the intervening objects no longer commute.

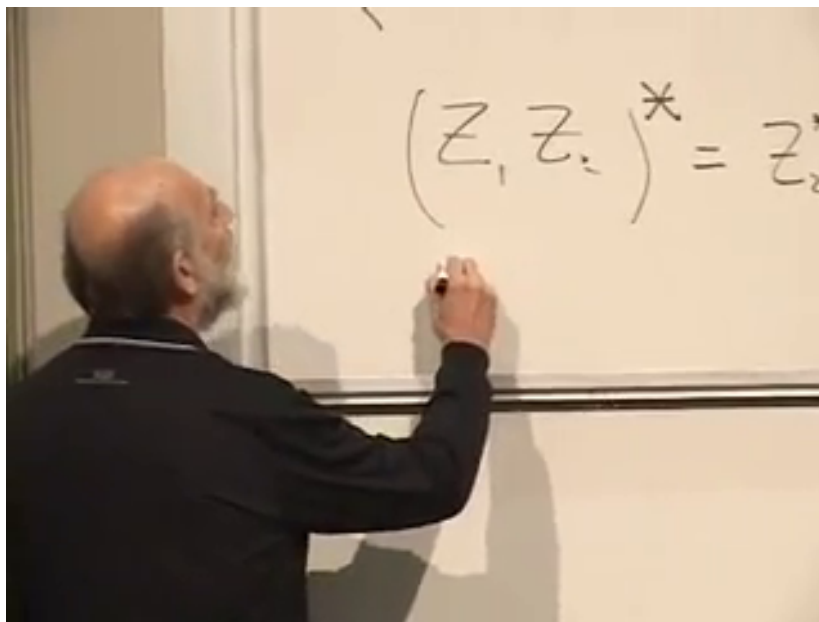


Figure 1.2: Complex conjugation of a product on the board. The corrected relation is given cleanly in Eqs. (1.4) and (1.5).

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Question from the audience. Can the complex number be written in exponential form? ^a

Response. Certainly. One may write $z = re^{i\theta}$, but we do not need that form yet. The Cartesian form makes conjugation and the coming vector-space rules visible with less machinery.

^aLecture timestamp: 00:15:12–00:15:18.

1.3 What Makes a Complex Vector Space?

A complex vector space is a collection of objects that may be added and multiplied by complex numbers without leaving the collection. If $|A\rangle$ and $|B\rangle$ are vectors and $\alpha, \beta \in \mathbb{C}$, then

$$\alpha|A\rangle + \beta|B\rangle \tag{1.6}$$

must be another vector in the same space. The vectors need not be arrows. What matters is the algebra of addition and scalar multiplication.

Several near-examples make the definition precise. Ordinary arrows form a real vector space, but not a complex vector space unless an additional rule for multiplication by i is supplied. The real numbers form a real vector space, not a complex one. The positive real numbers do not even form a real vector space: multiplying a positive number by -1 leaves the set.

1.3.1 Functions and Columns

Complex-valued functions of a real variable form a complex vector space. If $\psi(x)$ and $\phi(x)$ are such functions, then

$$(\alpha\psi + \beta\phi)(x) = \alpha\psi(x) + \beta\phi(x) \tag{1.7}$$

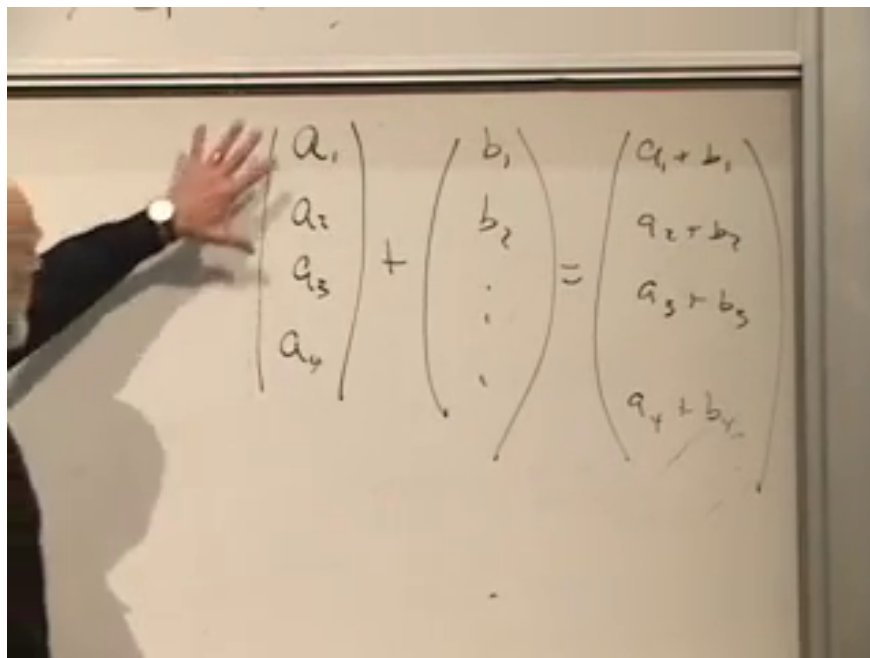


Figure 1.3: Componentwise addition of two column vectors. A column may be read as a function of a discrete index.

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is another complex-valued function.

A column vector is the same idea with a discrete argument. For four components,

$$|a\rangle = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix}, \quad |b\rangle = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{pmatrix}, \quad (1.8)$$

and

$$\alpha|a\rangle + \beta|b\rangle = \begin{pmatrix} \alpha a_1 + \beta b_1 \\ \alpha a_2 + \beta b_2 \\ \alpha a_3 + \beta b_3 \\ \alpha a_4 + \beta b_4 \end{pmatrix}. \quad (1.9)$$

The index 1, 2, 3, 4 plays the role of a discrete coordinate.

1.3.2 Dimension and Spanning

The dimension of a vector space is the minimum number of independent vectors needed to span it. One nonzero vector spans a line. Two noncollinear vectors span a plane: every vector $|v\rangle$ in that plane can be written

$$|v\rangle = r_1|v_1\rangle + r_2|v_2\rangle. \quad (1.10)$$

A third vector in the same plane adds no new direction, because it is already a linear combination of the first two. In three dimensions we need three noncoplanar vectors.

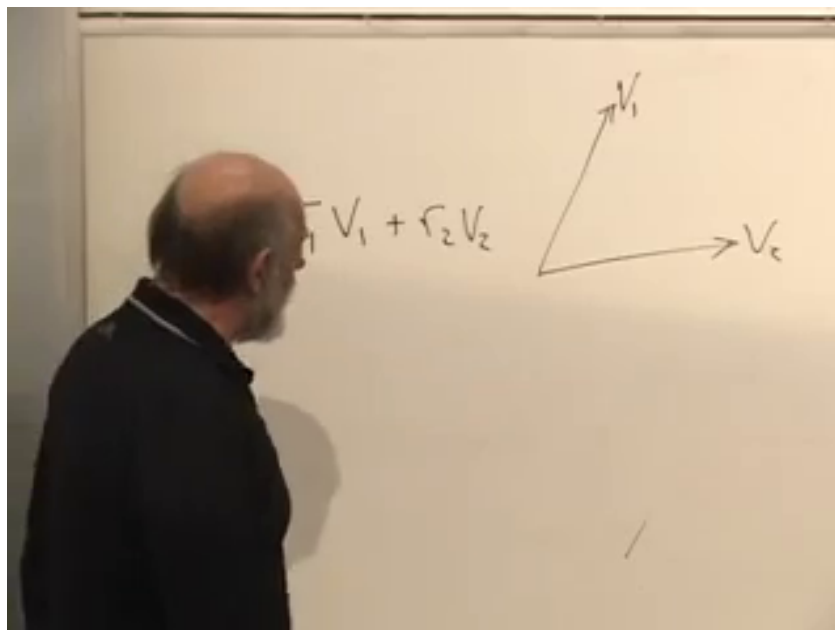


Figure 1.4: Two noncollinear vectors spanning a plane. The expression at left is partly obscured, while the geometric pair is clear.

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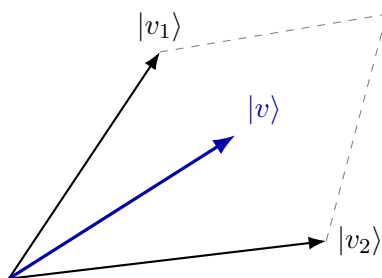


Figure 1.5: A clean reconstruction of the spanning argument: $|v\rangle = r_1|v_1\rangle + r_2|v_2\rangle$.

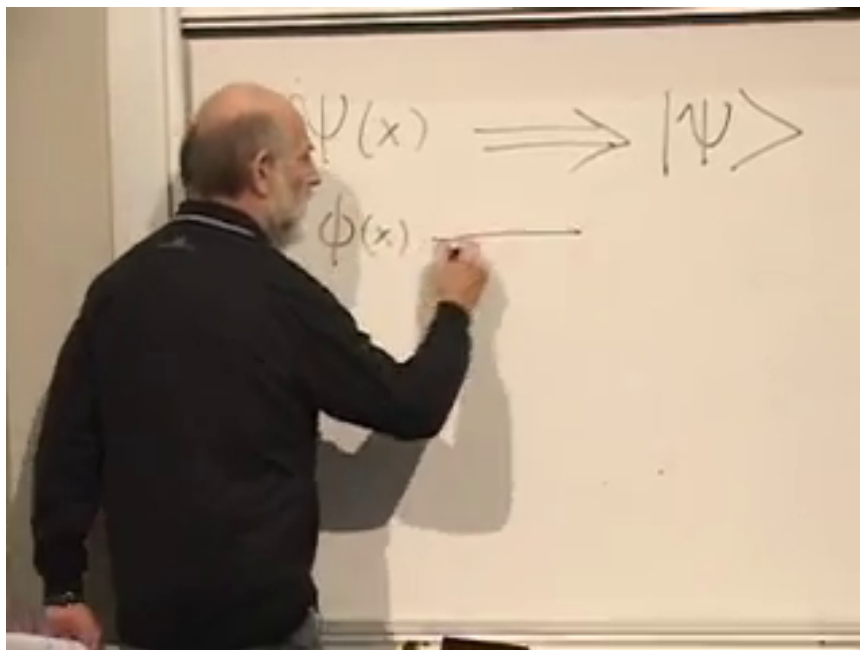


Figure 1.6: The board changes viewpoint from the function $\psi(x)$ to the abstract ket $|\psi\rangle$. The second line is still being written.

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The four-component columns form a four-dimensional space. A convenient spanning set is

$$\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \quad (1.11)$$

Remove the last column and no linear combination can produce a nonzero fourth component. By contrast, one-component columns are just complex numbers and form a one-dimensional complex space.

The space of smooth complex functions has no finite spanning set; it is infinite-dimensional. The details of bases in infinite-dimensional spaces will not concern us yet. For now, the essential point is that functions and columns are both concrete representations of abstract vectors:

$$\psi(x) \mapsto |\psi\rangle, \quad \begin{pmatrix} \psi_1 \\ \psi_2 \\ \vdots \end{pmatrix} \mapsto |\psi\rangle. \quad (1.12)$$

1.4 Bras, Kets, and the Dual Space

To every ket $|A\rangle$ we associate a bra $\langle A|$. A bra is a linear functional: when it acts on a ket, it produces a complex number. Once an inner product has been chosen, the Riesz correspondence identifies each ket with a bra antilinearly. In finite-dimensional coordinates this is the familiar

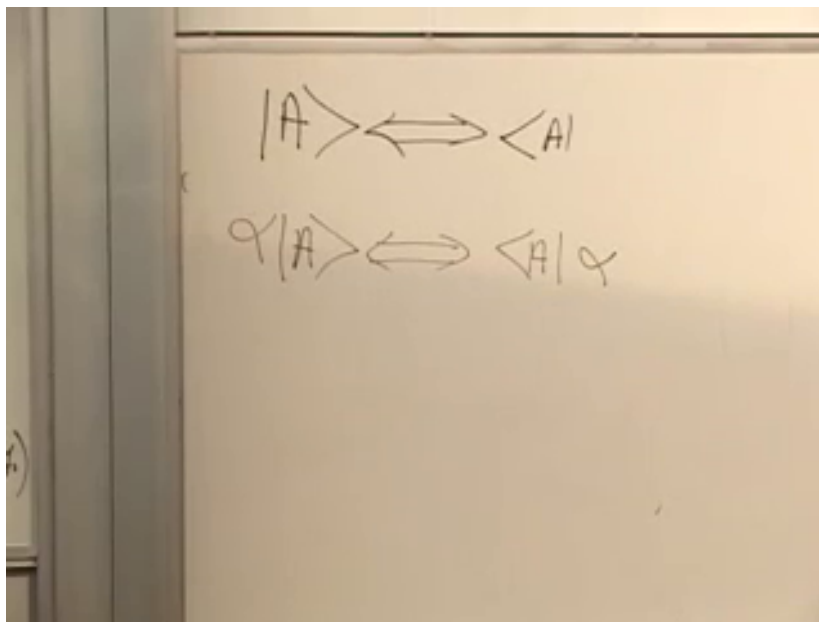


Figure 1.7: Ket-to-bra correspondence and the conjugation of a scalar coefficient.

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conjugate transpose:

$$|A\rangle = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_d \end{pmatrix} \longleftrightarrow \langle A| = \begin{pmatrix} a_1^* & a_2^* & \cdots & a_d^* \end{pmatrix}. \quad (1.13)$$

For a function, the corresponding bra is represented by the conjugate function $\psi^*(x)$.

The correspondence is antilinear:

$$\alpha|A\rangle \longleftrightarrow \langle A|\alpha^*, \quad (1.14)$$

$$|A\rangle + |B\rangle \longleftrightarrow \langle A| + \langle B|. \quad (1.15)$$

Question from the audience. Is there any significance to writing α^* on the right of the bra?^a**Response.** Not while α is an ordinary complex number, because scalar multiplication commutes. It is partly a convention for keeping formulas orderly. Position will matter when noncommuting operators enter the calculation.^aLecture timestamp: 00:43:24–00:43:57.

1.5 Inner Products

For ordinary real vectors, the dot product is

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta = \sum_i a_i b_i. \quad (1.16)$$

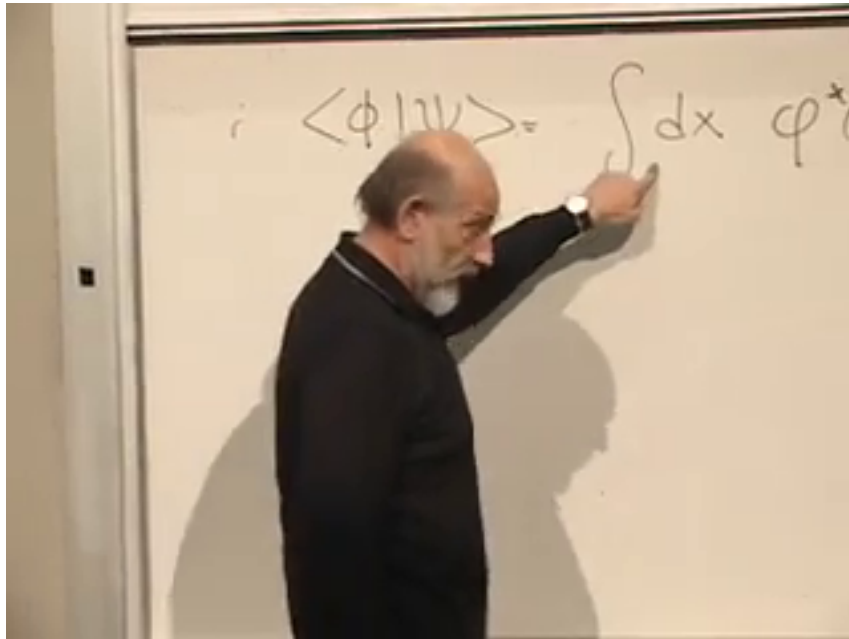


Figure 1.8: The function-space inner product being assembled on the board. The complete formula is Eq. (1.21).

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The inner product $\langle A|B \rangle$ is its complex counterpart. We use the convention standard in physics: it is linear in the ket and conjugate-linear in the bra. Thus

$$\langle A|(\beta B + \gamma C)\rangle = \beta \langle A|B\rangle + \gamma \langle A|C\rangle, \quad (1.17)$$

$$\langle \alpha A|B\rangle = \alpha^* \langle A|B\rangle. \quad (1.18)$$

It also obeys conjugate symmetry,

$$\langle A|B\rangle = \langle B|A\rangle^*. \quad (1.19)$$

Setting $B = A$ shows that $\langle A|A \rangle$ is real. Positivity is an additional axiom:

$$\langle A|A \rangle \geq 0, \quad \langle A|A \rangle = 0 \iff |A\rangle = 0. \quad (1.20)$$

It is not a consequence of conjugate symmetry alone. Relativity supplies a useful warning: the Minkowski metric admits positive, negative, and null norms, so it is not an inner product in this positive-definite sense.

For complex-valued functions on a real interval I ,

$$\langle \phi | \psi \rangle = \int_I dx \phi^*(x) \psi(x). \quad (1.21)$$

For columns,

$$|a\rangle = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_d \end{pmatrix}, \quad \langle b| = \begin{pmatrix} b_1^* & b_2^* & \cdots & b_d^* \end{pmatrix}, \quad (1.22)$$

and therefore

$$\langle b|a\rangle = \sum_{i=1}^d b_i^* a_i. \quad (1.23)$$

Question from the audience. Are the coordinates of the vector b the starred quantities?^a

Response. The ket $|b\rangle$ has coordinates b_i . The corresponding bra $\langle b|$ has entries b_i^* . The bra symbol already indicates that conjugation has been made; one does not rename the ket coordinates themselves.

^aLecture timestamp: 00:57:55–00:58:37.

Question from the audience. Must the variable x in the function inner product be real?^a

Response. In the present construction, yes. We may use several real variables, say x and y , and integrate over each of them. Functions of complex arguments can also be studied, but they are not needed for this course at this point.

^aLecture timestamp: 00:58:59–00:59:42.

1.6 Orthonormal Bases and Components

Choose d vectors $\{|b_i\rangle\}_{i=1}^d$ in a d -dimensional space. They are normalized when $\langle b_i|b_i\rangle = 1$, with no sum on i , and mutually orthogonal when $\langle b_i|b_j\rangle = 0$ for $i \neq j$. Both statements are contained in

$$\langle b_i|b_j\rangle = \delta_{ij}, \quad \delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases} \quad (1.24)$$

An orthonormal set of d vectors in a d -dimensional space is a basis. The columns in Eq. (1.11) provide the standard example. Their corresponding bras are rows. A blackboard slip between the row and column forms was caught in the room; the corrected notation is the one used here.

Any vector can be expanded in the basis:

$$|v\rangle = \sum_{i=1}^d v_i |b_i\rangle. \quad (1.25)$$

Multiply from the left by $\langle b_j|$:

$$\langle b_j|v\rangle = \sum_{i=1}^d v_i \langle b_j|b_i\rangle \quad (1.26)$$

$$= \sum_{i=1}^d v_i \delta_{ji} = v_j. \quad (1.27)$$

The component v_j is the inner product with the j -th basis vector. Substitution gives the form we shall use repeatedly,

$$|v\rangle = \sum_{i=1}^d |b_i\rangle \langle b_i|v\rangle. \quad (1.28)$$

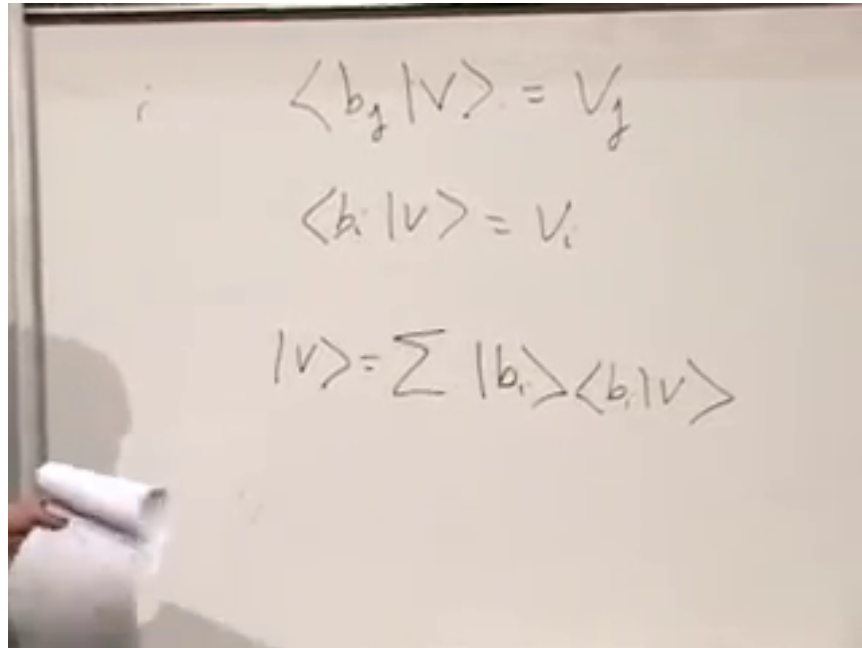


Figure 1.9: Extraction of a basis component and reconstruction of $|v\rangle$. The board preserves the same sequence as Eqs. (1.27) and (1.28).

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1.7 Classical Alternatives and Quantum Superpositions

We can now attach physical meaning to the vector space. A classical phase space is a set of possible states. Subsets support the ordinary logic of union and intersection, OR and AND. Quantum states carry additional linear structure. Adding two state vectors and multiplying them by complex numbers can produce another physical state; mere set membership cannot express this.

Begin with a coin. Its classically distinguishable configurations are heads and tails. We may call them H and T , or label them $+1$ and -1 , but those labels are names, not vector lengths. The quantum model assigns orthonormal states

$$\langle H|H\rangle = \langle T|T\rangle = 1, \quad \langle H|T\rangle = 0. \quad (1.29)$$

The real physical example behind the coin is an electron with two distinguishable spin outcomes.

The orthogonality has operational content. Configurations that one ideal measurement can distinguish with certainty are represented by orthogonal vectors. This is one of the postulates we shall use. For a six-sided die, the same rule gives six states $|1\rangle, \dots, |6\rangle$ satisfying

$$\langle i|j\rangle = \delta_{ij}. \quad (1.30)$$

Now comes the step that has no classical analogue:

$$|C\rangle = \alpha_H |H\rangle + \alpha_T |T\rangle. \quad (1.31)$$

This 'confused coin' is a genuine quantum state, neither heads nor tails. It is a coherent superposition,

not a classical statistical mixture whose member secretly has one definite face. Likewise,

$$|D\rangle = \sum_{i=1}^6 \alpha_i |i\rangle \quad (1.32)$$

is a superposition of the six die outcomes.

1.8 Amplitudes, Probabilities, and Normalization

Question from the audience. How did the coefficients α_H and α_T become probabilities?^a

Response. They do not themselves become probabilities. They are probability amplitudes, and the Born rule is a postulate:

$$P(H) = |\alpha_H|^2, \quad P(T) = |\alpha_T|^2.$$

This is not the whole meaning of the amplitudes; their phases contain more information than these two probabilities reveal. Operationally, prepare a large ensemble of systems in the same state, make the same measurement on each, and identify the probabilities with the limiting frequencies of heads and tails.

^aLecture timestamp: 01:33:59–01:37:49.

If all outcomes are accounted for, their probabilities add to one:

$$|\alpha_H|^2 + |\alpha_T|^2 = 1. \quad (1.33)$$

Using Eq. (1.29), this is precisely

$$\langle C|C\rangle = 1. \quad (1.34)$$

For the die,

$$\sum_{i=1}^6 |\alpha_i|^2 = 1, \quad \langle D|D\rangle = 1. \quad (1.35)$$

Physical state vectors are therefore normalized.

Question from the audience. For equal heads and tails probabilities, are both amplitudes $1/\sqrt{2}$?^a

Response. Their magnitudes are $1/\sqrt{2}$, but the amplitudes may be complex:

$$\alpha_H = \frac{e^{i\theta_H}}{\sqrt{2}}, \quad \alpha_T = \frac{e^{i\theta_T}}{\sqrt{2}}.$$

Every complex number on the unit circle has unit magnitude. Multiplying such a number by $1/\sqrt{2}$ gives probability $1/2$. For six equally likely die outcomes, each amplitude has magnitude $1/\sqrt{6}$.

^aLecture timestamp: 01:38:02–01:39:24.

We have not yet said how to prepare an electron in the analogue of this half-heads, half-tails state. That experimental question will return when spin replaces the toy coin. For the moment it is more efficient to state the rules, learn how they work, and then ask which experiments forced physicists to adopt them. The first rules are now visible:

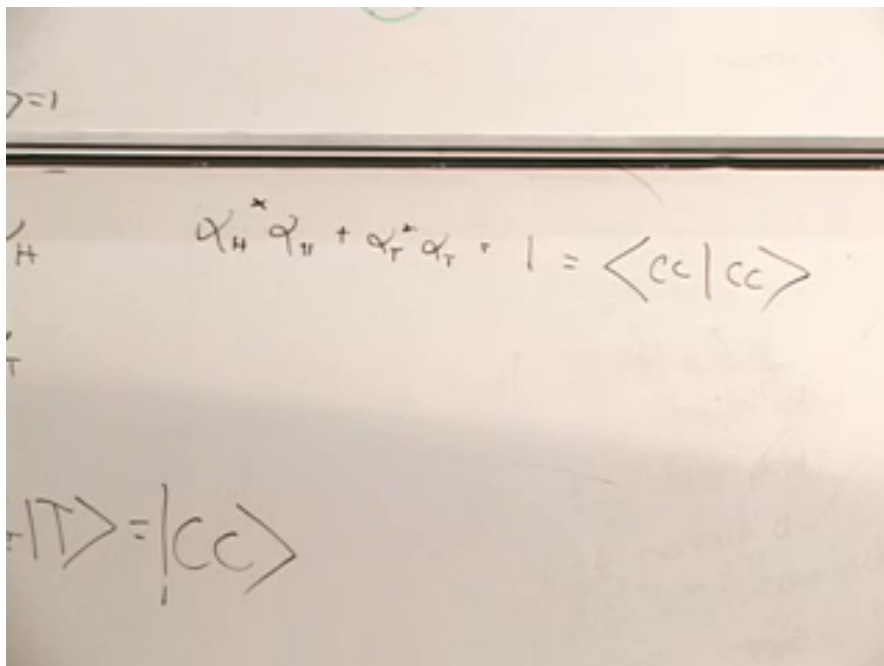


Figure 1.10: Normalization of the two-state superposition written as both a sum of squared amplitudes and the norm $\langle C|C \rangle$.

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1. states are represented by vectors in a complex vector space;
2. perfectly distinguishable alternatives are represented by orthogonal vectors;
3. physical state vectors have unit norm; and
4. squared amplitude magnitudes give probabilities in an orthonormal measurement basis.

1.9 The First Look at Operators

There is one more mathematical object to introduce. An operation assigns a new vector to every vector in the space. Fixed-angle rotation, uniform rescaling, and reflection about an axis are familiar examples. Squaring a vector's length is also an operation, but it is not linear: a vector of length 2 is sent to one of length 4, while a vector of length 3 is sent to one of length 9.

We denote a linear operator by a hatted letter, such as $\hat{L}$. It obeys homogeneity and additivity:

$$\hat{L}(\alpha|A\rangle) = \alpha \hat{L}|A\rangle, \quad (1.36)$$

$$\hat{L}(|A\rangle + |B\rangle) = \hat{L}|A\rangle + \hat{L}|B\rangle. \quad (1.37)$$

Equivalently, for arbitrary complex α, β ,

$$\hat{L}(\alpha|A\rangle + \beta|B\rangle) = \alpha \hat{L}|A\rangle + \beta \hat{L}|B\rangle. \quad (1.38)$$

Rotation by a fixed angle passes both tests. Rotating twice a vector gives twice the rotated vector. If $\mathbf{a} + \mathbf{b}$ is the diagonal of a parallelogram, rotating the whole parallelogram shows

$$R(\mathbf{a} + \mathbf{b}) = R\mathbf{a} + R\mathbf{b}. \quad (1.39)$$

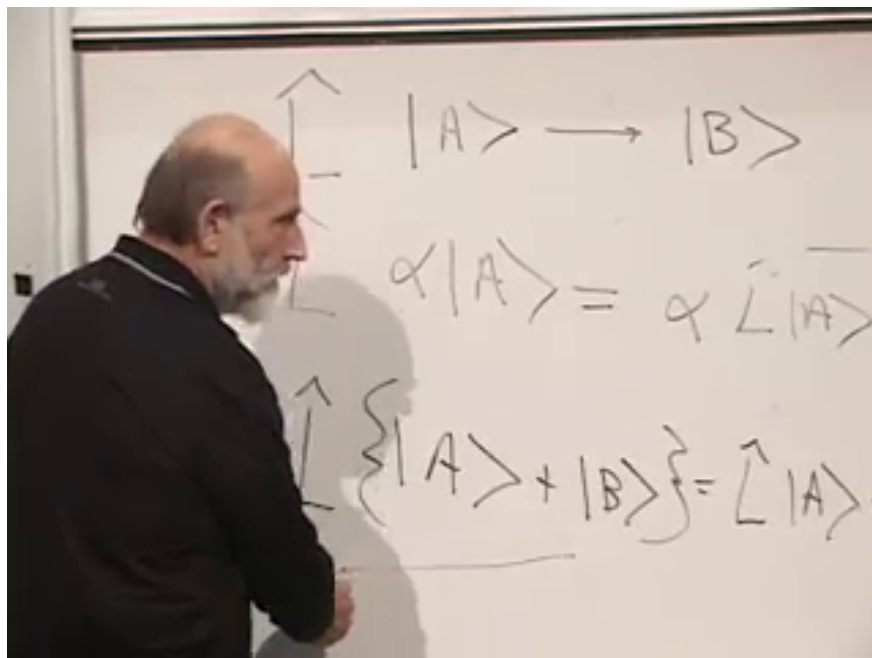


Figure 1.11: The two defining rules for a linear operator: scalar multiplication passes through the operator, and the image of a sum is the sum of the images.

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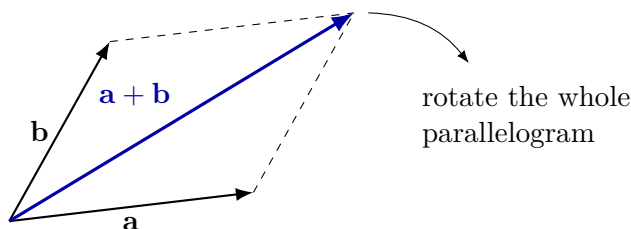


Figure 1.12: The parallelogram proof of additivity for a fixed rotation.

Uniform doubling and reflection are linear for the same reason. The length-squaring operation fails: scaling the input by α changes the length squared by $|\alpha|^2$, not by α .

State vectors describe the possible states of a quantum system. Linear operators will describe measurable quantities, or observables: the quantum version of asking whether the coin is heads or tails. The precise relation between an observable, its operator, and the outcomes of a measurement is the next step. We are now close enough to the formalism to begin making predictions.